*BY ORDER OF THE COMMANDER*
*TYNDALL AIR FORCE BASE*

*TYNDALL AIR FORCE BASE*
*INSTRUCTION 15-101*

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***Weather***

***WEATHER SUPPORT***

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This instruction implements Department of the Air Force Manual (DAFMAN) 15-129, *Air and Space Weather Operations*. It establishes responsibilities and weather support procedures, general information including weather services to include weather observations and forecasts, weather watches, warnings, and advisories (WWAs), space weather, supported services, dissemination of information, and reciprocal support. See **Attachment 1** for a Glossary of Reference and Supporting Information. This publication applies to units assigned to the 325th Fighter Wing (FW), subordinate units, and units assigned or attached to, or supported by Tyndall Air Force Base (TAFB). Refer recommended changes and questions about this publication to the OPR using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. The authorities to waive wing, unit, delta or garrison level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority or alternately to the publication OPR for non-tired compliance items. See DAF Manual (DAFMAN) 90-161, *Publishing Processes and Procedures,* for a description of the authorities associated with the tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval, authority, or alternately, to the Publication OPR for non- tiered compliance items. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the AF.

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# Chapter 1

# GENERAL INFORMATION

## 1.1. Overview.

1.1.1. The 325th Operations Support Squadron Weather Flight (WF) and the 26th Operational Weather Squadron (26 OWS) are the official weather information agencies for Tyndall Air Force Base (TAFB) and associated tenant units. The WF and 26 OWS also provide support to other tenant and transient units which do not have other means of obtaining weather support.

## 1.2. Designators.

1.2.1. All weather products disseminated by or for TAFB will use KPAM as the International Civil Aviation Organization location indicator. The World Meteorological Organization (WMO) index number for KPAM is 747750. Additionally, a fixed meteorological equipment (FMQ) weather sensor located on the drone runway approximately 4 miles southeast from the primary KPAM runway complex is designated as KTDR.

## 1.3. Responsibilities.

1.3.1. General responsibilities of the 26 OWS and WF are outlined in AFI 15-128 and DAFMAN 15-129. Specific responsibilities of the 26 OWS and the WF are defined in the Installation Data Page at **https://26ows.us.af.mil/tech_ref/idp/?usehf=1&icao=KPAM**. Additionally, the WF will support 325 FW and tenant units as directed by the 325 FW/CC.

## 1.4. Customer Weather Limitations.

1.4.1. 325 FW and tenant unit customer weather limitations/impacts can be found in **Attachment 2**.

## 1.5. Duty Priorities.

1.5.1. The WF will provide weather support IAW the duty priority list found in **Attachment 3**.

## 1.6. Operating Hours and Contact Information.

1.6.1. The WF offers weather services 24/7, 365 days a year as specified in the TAFB Flight Information Publication (FLIP). WF Ops Desk Defense Switched Network (DSN): 523-4235/4203, Commercial: (850) 283-4235/4203. WF Standby Commercial: (850) 381-1956. WF Leadership can be reached at 4200/4205.

1.6.2. The 26 OWS provides weather support to TAFB and transient aircrews if the WF is unable to conduct operations. 26 OWS DSN: 331-2634 or Commercial: 318-529-2634.

## 1.7. Access to Weather Information.

1.7.1. Primary websites to access weather information for Tyndall are:

1.7.1.1. Joint Environmental Toolkit (JET) Guest Portal: **https://owsjet26.us.af.mil**

1.7.1.2. OSW SharePoint: **https://usaf.dps.mil/sites/325OG/OSS/OSW/SitePages/**Home.aspx

1.7.1.3. 26 OWS Web Page: **https://26ows.us.af.mil/**

**1.8. Backup Weather Support.**

1.8.1. WF Alternate (Back-up) Operating Location. In the event of a building evacuation, forecasters will relocate to the 7th floor of the Air Traffic Control (ATC) Tower (bldg 220, 325 OSS/OSAT) to continue providing weather services for TAFB operations and to perform the WF's "eyes forward" responsibility for the 26 OWS. See **Attachment 4** for WF and ATC Tower procedures during WF AOL operations.

1.8.2. Back-up Communication/Weather Equipment. In the event primary weather sensing equipment connectivity is lost, back-up equipment will be used, as necessary. WF weather services will be degraded (weather products, pilot briefings, etc) due to limited facilities and loss of dedicated data services, including satellite and radar feeds.

**1.9. Release of Weather Information to Non-Department of Defense Agencies and Individuals.**

1.9.1. Weather information will not be released to non-Department of Defense agencies or the general public without approval from the Wing Public Affairs and Legal offices. Any questions or requests for weather information must be coordinated through the WF Commander and/or Flight Chief.

# Chapter 2

# AIRFIELD SERVICES ELEMENT

## 2.1. Airfield Weather Observations.

2.1.1. Observations are taken, recorded, and disseminated by the WF IAW AFMAN 15-111 utilizing AF IMT 3803within the JET dissemination system. See **Attachment 4** for further details on Tyndall special observation criteria.

2.1.1.1. FMQ-19. The FMQ-19 is the certified AF fixed base weather observing system (FBWOS) responsible for taking automated Routine Meteorological Observation Report (METAR) and Special (SPECI) observations. The FMQ-19 system consists of two sensor sites; the primary is located at the northwest end of the airfield while the secondary sensor is on the southeast end.

2.1.2. Maintenance Mode. Whenever a sensor is inoperative the automated system will set a "$" after an observation. The "$" represents a failure message due to insufficient or corrupted data from the sensor. WF personnel must inform the Radar Airfield & Weather Systems (RAWS) flight of weather sensor equipment failure. RAWS will then log in as a "technician" to remove any failure message(s).

2.1.3. Official Observing Site. This is the physical location of the FMQ-19 observation. The primary sensor suite is located at the northwest end of the airfield on the northeast side of the right and left runway. In the event that back-up of the FMQ-19 is required, the official observation location moves to near the storm drain on the northeast side of building 290.

## 2.2. Terminal Aerodrome Forecast (TAF).

2.2.1. Tyndall TAFs will be produced and disseminated by the WF IAW AFI 15-128, AFMAN 15-124, AFMAN 15-129, and as outlined in the Installation Data Page. TAFs are valid for a 30-hour period and apply to the area within a 5-nautical mile (NM) radius of the designated center point of the Tyndall Airfield. TAFs are issued every 8 hours.

2.2.2. Specification and Amendment Criteria. The TAF Specification and Amendment criteria are listed in **Attachment 5** and are specified IAW DAFMAN 15-129.

2.2.3. Dissemination. The WF is responsible to prepare and disseminate the TAF via JET. If JET is inoperative, TAFs will be disseminated by the WF to ATC facilities and the Supervisor of Flying (SOF) during flying hours.

## 2.3. Augmenting Observations.

2.3.1. WF personnel will augment observations from the FMQ-19 when augmentation is required IAW AFMAN 15-111. During augmented observations, WF personnel will recheck weather conditions, at 20 minutes intervals, when any of the following conditions are observed to be occurring or are forecast to occur within 1 hour.

2.3.2. Ceilings form below, decrease to less than or dissipate/increase to or exceed 2,000 ft.

2.3.3. Visibility decreases to less than, increases to equal, or exceeds 3 statute miles (SM) (4,800 meters).

2.3.4. Precipitation (any form).

2.3.5. Thunderstorms and hail.

2.3.6. Fog or Mist.

2.3.7. Tornado, waterspout, or funnel cloud.

2.3.8. Freezing precipitation or ice pellets.

2.3.9. Volcanic ash.

2.3.10. Tower visibility remark (only during controlled airfield hours).

**2.4. Cooperative Weather Watch.**

2.4.1. This is the process by which Air Traffic Control (ATC) personnel, the Supervisor of Flying (SOF), flying units, and Security Forces personnel report changes in weather conditions to the duty weather technician. These inputs can be utilized for issuing, updating, or amending various weather products. Examples of Cooperative Weather Watch items are listed below.

2.4.1.1. Procedures for task certified ATC personnel to report changes in tower visibility when it is less than 4 SM (6000 m) and differs from the prevailing visibility by at least one reportable value.

2.4.1.2. Precipitation begins or ends (tower and flying units primarily).

2.4.1.3. Thunderstorms and/or lightning.

2.4.1.4. Sea fog.

2.4.1.5. Waterspouts or tornadoes.

2.4.1.6. Any other meteorological condition they consider significant (hail, wind damage, etc.).

2.4.1.7. Procedures for ATC personnel to relay Pilot Reports (PIREP) as soon as practical, within ATC established duty priorities.

**2.5. Meteorological Watch (METWATCH).**

2.5.1. METWATCH is a deliberate, continuous process for monitoring terrestrial weather or the space environment in an area or region. The purpose of METWATCH is to identify when and where observed conditions significantly diverge from forecast conditions, determine courses of action to update or amend a forecast product or group of products, and notify designated agencies.

**2.6. Fighter Index of Thermal Stress (FITS).**

2.6.1. The WF determines the FITS condition IAW AFI 48-151. An observed weather advisory will be issued when the FITS caution or danger are reached and cancelled when these conditions no longer exist. Additionally, WF personnel will contact the Command Post via phone when FITS conditions change. FITS advisories are only issued during hours in which the airfield is open.

**2.7. Pilot to Metro Service (PMSV).**

2.7.1. The WF currently does not have a functioning PMSV radio. However, the WF does maintain the ability to provide weather via phone patches from Command Post.

**2.8. FBWOS Sensor Selection.**

2.8.1. Upon notification from ATC that a runway change has been completed, the WF forecaster will change the appropriate weather sensors to match the active runway.

**2.9. Flight Information Publication (FLIP) Weather Updates.**

2.9.1. The WF is responsible for ensuring all weather information, such as WF operating hours and meteorological equipment limitations, in the FLIP is accurate. The WF will route all weather related updates through the Airfield Management FLIP Manager.

**2.10. Mishap Procedures.**

2.10.1. Encode & disseminate a full element SPECI observation IAW AFMAN 15-111, regardless of the operating mode. NOTE: A SPECI is not required for an in-flight emergency (IFE); however, this should alert weather personnel to be prepared to take a SPECI if the IFE becomes a mishap.

2.10.2. Save applicable terrestrial and space weather data when notified of any aircraft mishaps (weather-related or not) requiring OPREP-3 reporting and coordinate with other units (e.g. OWS, TACC) to ensure required data is saved.

2.10.3. Ensure all data used in the development of any weather information, product, or service provided to a customer are saved, including but not limited to: Meteorological Satellite imagery, RADAR products, upper air package, hazard charts, PIREPs, Significant Meteorological Information, TAFs, observations, Watch, Warning, and Advisory (WWA), upper air soundings, and briefing material provided.

2.10.4. The WF will notify the OWS Senior Duty Officer of all aircraft mishaps as soon as possible after notification of the event.

2.10.5. Gather the saved data and hold it until requested or send the data to the organization requesting the data save (accident investigation board, safety investigation board, etc.). The WF will not dispose of the data until the organization initiating the data save determines it is no longer needed and coordinates the disposition with all organizations involved.

# Chapter 3

# MISSION INTEGRATION FUNCTION

## 3.1. General.

3.1.1. Mission services are those actions directly related to completing each supported units' mission(s) (e.g. Mission Weather Products (MWP), Mission Briefs, and Flight Weather Briefings). Understanding operational tactics, weapons platforms capabilities and sensitivities is pivotal to injecting timely, accurate and relevant environmental intelligence at every decision point in the mission planning and execution process.

## 3.2. Operating Areas.

3.2.1. Detailed maps of the operating areas used by TAFB are available to WF members in the WF's Forecaster Reference Notebook, as well as visual aids in the work area. WF forecasters will be familiar with these areas.

## 3.3. Mission Weather Limitations.

3.3.1. A detailed list of mission profiles and associated weather limitations by each flying unit on TAFB will be coordinated and documented annually. These can be found in **Attachment 2**.

## 3.4. Mission Weather Product (MWP).

3.4.1. The MWP is a mission-specific forecast developed by the WF using the Mission Execution Forecast Process outlined in DAFMAN 15-129. During this process, the WF fuses and tailors products created by strategic and theater weather centers (e.g., TAFs, hazard bulletins, etc.), as well as information supplied by local units and supporting agencies. Due to the lack of weather data available in most of the local Mission Operating Area (MOA), PIREPs play a very important role in keeping the MWP current and accurate. The MWP must be horizontally consistent with (but not necessarily mirror) products issued by the respective OWS.

3.4.2. The Tyndall MWP is available through the Tyndall WF SharePoint at **https://usaf.dps.mil/sites/325OG/OSS/OSW/SitePages/Home.aspx**. In the event of a Local Area Network (LAN) outage, the MWP will be made available for pick-up at the weather desk. MWPs are not issued on weekends, holidays, or down days when the airfield is closed without prior coordination with WF leadership.

3.4.3. MWP Amendment Criteria. The Working Area portions of the Web-Based MWP are amendable. Amendments deviating from published OWS charts must be forwarded to the OWS for consideration into amending their products. These areas will be amended anytime the thresholds defined below are crossed:

3.4.3.1. KPAM ceiling heights greater than or equal to (GTE)/less than (LT) 002, 003, 005, 007, 010, 015, 020.

3.4.3.2. KPAM visibility (statue miles) GTE/LT 1, 1.5, 2, 3.

3.4.3.3. KPAM crosswinds GTE 20 kts.

3.4.3.4. KPAM observed/forecasted rain or thunderstorms.

3.4.3.5. Surface winds (over land) are observed GTE 30 kts.

3.4.3.6. Thunderstorms within a MOA are forecast or observed to be scattered in coverage.

3.4.3.7. Moderate or greater turbulence.

3.4.3.8. Light or greater icing.

3.4.3.9. 13K clear airspace between 050 and 600.

3.4.3.10. 1 NM in-flight visibility.

3.4.3.11. 5 NM in-flight visibility.

3.4.3.12. Cloud forecast changes below 5K.

**3.5. Mission Planning Products.**

3.5.1. The WF produces routine planning weather products that can be found on the Tyndall Weather SharePoint site as well as the 26 OWS KPAM Unit Tailored Page **https://26ows.us.af.mil/tech_ref/utp/index_utp.cfm?utp_id=44**. Any additionally planning weather product creation should be coordinated with WF leadership via email at **325oss.weatherflight@us.af.mil**.

**3.6. Flight Weather Briefings.**

3.6.1. Tyndall weather personnel will provide verbal or traditional Flight Weather Briefings (DD Form 175-1) to aircrews as requested following the flight's duty priorities. Briefings will be given at the weather desk, via e-mail, or over the phone. Aircrews will provide weather personnel as much lead-time as possible (at least 2 hours) to complete the request by calling the WF.

3.6.2. Mass briefings for special missions require 48-hours advanced notice and are subject to staffing availability.

3.6.3. Transient aircrews will be provided with an original briefing or be given an update to an existing briefing form as time and resources allow. If unable to provide a briefing or update an existing briefing, the aircrew will be directed to the 26 OWS and given contact information.

3.6.4. Off-station Support. IAW DAFMAN 15-129 "provide or arrange for" concept of operations, Tyndall WF is the primary source for tailored weather information for 325 FW units. When mission activities occur away from the main operating location, Tyndall WF leadership will assume responsibility to determine the most effective means of ensuring their units receive mission execution weather information.

**3.7. MISSIONWATCH.**

3.7.1. This is the process by which the WF monitors the weather for all local missions, to include all working areas. All on-site meteorological and commercial data sources are leveraged to accomplish this task. During rapidly changing weather, the WF will amend/update the MWP as required and contact the SOF to pass on critical changes.

**3.8. Sea Conditions.**

3.8.1. The WF will notify the Command Post, and SOF when any sea condition advisory for the over-water working areas is issued. Sea Conditions are listed in **Table 3.1**. Sea Condition Red and Sea Temperature Red are defined IAW AFI 11-202V3 ACC Supp. and AFI 11-301V1 ACC Supp.

**Table 3.1. Sea Conditions.**

| Threat Level | Threshold |
|---|---|
| Condition Yellow | Waves ≥ 8FT and/or sustained SFC winds ≥ 20kts |
| Condition Red | Waves ≥ 10FT and/or sustained SFC winds ≥ 25kts |
| Temperature Red | Water temperature ≤ 60ºF/15.5ºC |

**Note:** Sea temperature red conditions are annotated on the MWP but the WF does not issue an observed advisory for this condition.

### 3.9. Post-Mission Analysis/Feedback.

3.9.1. Per DAFMAN 15-129, the WF will develop internal processes to improve the MWP based on feedback from the supported units. Flying units that regularly utilize weather support from the Tyndall WF should provide post-mission/utilization feedback, when feasible. Formal/informal feedback methods include:

3.9.1.1. Email and/or phone calls to the WF leadership.

3.9.1.2. Face-to-face feedback.

3.9.1.3. Post-mission debrief.

3.9.1.4. PIREP via Phone patch.

### 3.10. Space Weather Support and Services.

3.10.1. Many weapons and communications systems use satellites and radio waves (High Frequency (HF), Very High Frequency (VHF), Ultra High Frequency (UHF), and Satellite Communications (SATCOM), which can be rendered useless by electro-magnetic radiation from the sun. Most Space Weather products are very short-term forecasts updated at 6-hour intervals. For more information concerning these products, contact the WF or utilize the AF Space Weather Main Page at: **https://weather.af.mil/confluence/display/AFWWEBSTBT/Space+Weather+Main+Page**

# Chapter 4

# STAFF WEATHER ELEMENT

## 4.1. General.

4.1.1. Staff services are those briefings provided primarily by WF leadership. These briefings are specialized and focused on a particular event/audience. Some examples are tropical bulletins, 325 FW Stand-Up, climatology data and briefings, and Air Control Squadron support.

## 4.2. Five Day Forecast.

4.2.1. The WF will produce a 5-day forecast outlining potential weather impacts for the next five days. Each day is split into an a.m. and p.m. forecast, along with the forecasted high and low temperature. Mission impacts to aviation and personnel are depicted at the bottom. This product is produced Monday – Friday and can be produced on weekends by request. The 5-day forecast is not amendable and should be used for planning purposes only.

## 4.3. Climatology Data.

4.3.1. The WF tracks climate statistics for the base and leverages the support of the 14th Weather Squadron for climate statistics all over the world. Upon request from an authorized agency (as deemed appropriate by station management), the WF will provide climatology data for Tyndall or other locations.

## Chapter 5

## RESOURCE PROTECTION

### 5.1. General.

5.1.1. The WF is responsible for issuing all observed and forecasted watches, warnings, and advisories for TAFB.

### 5.2. Weather Watches.

5.2.1. A weather watch is a special notice to inform operational commanders of a potential for environmental conditions of such intensity as to pose a hazard to life or property (e.g. tornadoes, hail, etc.) and are used by operational commanders to make force protection and risk management decisions. Watches are issued for a 5 NM (10 NM for tornado) radius of the center-point of the Tyndall runway complex. The following table contains all of the weather watches and desired lead-times issued for Tyndall:

**Table 5.1. Watch Criteria.**

| Watch Type | Criteria | Desired Lead Time |
|---|---|---|
| Tornado (SWAP criteria) | Tornado, Water Spout, or Funnel Cloud (detected by RADAR or visually observed) AND threatening warned location / aerodrome complex | 75 minutes |
| Severe Thunderstorm (SWAP criteria) | Potential for Damaging Winds ≥ 50 kts associated with thunderstorms and/or Damaging Hail ≥ 3/4 inch at Tyndall AFB | 60 minutes prior to warning |
| Moderate Thunderstorm | Potential for Strong Winds ≥ 35 kts but < 50 kts associated with thunderstorms and/or Large Hail ≥ 1/4 inch but < 3/4 inch at Tyndall AFB | 60 minutes prior to warning |
| Damaging Winds (SWAP criteria) | Potential for Surface winds not associated with thunderstorms ≥ 50 kts at Tyndall AFB | 60 minutes prior to warning |
| Strong Winds | Potential for Surface winds not associated with thunderstorms ≥ 35 kts but < 50 kts at Tyndall AFB | 60 minutes prior to warning |
| Heavy Rain | Potential for Heavy Rain > 2 inches within 1 hr at Tyndall AFB | 60 minutes prior to warning |
| Heavy Snow | Potential for Heavy Snow > 1 inches within 1 hr at Tyndall AFB | 60 minutes prior to warning |
| Freezing Precipitation (SWAP Criteria) | Potential for Freezing Precipitation (Any Intensity) at Tyndall AFB | 120 minutes prior to warning |

## 5.3. Weather Warnings.

5.3.1. Weather warnings are a special notice to inform operational commanders when an established weather condition of such intensity as to pose a hazard to life or property is occurring or is expected to occur. Weather warnings provide concise information outlining environmental threats and are used by operational commanders to make force protection decisions. Warnings are issued for a 5 NM (10 NM for tornado) radius of the center-point of the Tyndall runway complex. Severe Weather Action Procedures (SWAP) will be used for warnings specified in **Table 5.2**.

**Table 5.2. Warning Criteria.**

| Warning Type | Criteria | Desired Lead Time |
|---|---|---|
| Tornado (SWAP criteria) | Tornado, Waterspout, or Funnel Cloud (detected by RADAR or visually observed) AND threatening warned location / aerodrome complex | 15 Minutes |
| Severe Thunderstorm (SWAP criteria) | Damaging Winds ≥ 50 kts associated with thunderstorms and/or Damaging Hail ≥ 3/4 inch at Tyndall AFB | 60 minutes |
| Moderate Thunderstorm | Strong Winds ≥ 35 kts but < 50 kts associated with thunderstorms and/or Large Hail ≥ 1/4 inch but < 3/4 inch at Tyndall AFB | 60 minutes |
| Damaging Winds (SWAP criteria) | Surface winds not associated with thunderstorms ≥ 50 kts at Tyndall AFB | 60 minutes |
| Strong Winds | Surface winds not associated with thunderstorms ≥ 35 kts but < 50 kts at Tyndall AFB | 60 minutes |
| Heavy Rain | Heavy Rain > 2 inches within 1 hr at Tyndall AFB | 60 minutes |
| Heavy Snow | Heavy Snow > 1 inches within 1 hr at Tyndall AFB | 90 minutes |
| Freezing Precipitation (SWAP criteria) | Freezing Precipitation (Any Intensity) at Tyndall AFB | 60 minutes |
| Freeze Warning | Freeze Warning (Temp < 32F for > 2 hrs) at Tyndall AFB | 12 hours |
| Lightning | Lightning within 5 NM from Tyndall AFB and/or Silver Flag | Observed |

5.3.2. Observed Weather Warnings. Lightning warnings are the only type of observed warning issued on Tyndall. Lightning warnings are not issued until lightning is observed, either audibly, visually, on the FMQ-19, through PIREP, radar interpretation, or via the National Lightning Detection Network, and remain valid until lightning has not occurred in the area for at least 15 minutes. A lightning warning will not be cancelled, however, if another thunderstorm within 15 NM is moving into the area.

## 5.4. Weather Advisories.

5.4.1. Weather advisories are a special weather product to alert an end user of the occurrence of, or imminent occurrence of weather conditions impacting operations. An observed weather advisory is a special notice sent to customers alerting them that a predefined weather phenomenon, which may impact operations, is occurring on TAFB or at specific mission airspaces. The majority of weather advisories for TAFB are observed advisories. Weather advisories are issued for:

**Table 5.3. Advisory Criteria.**

| Criteria | Forecast/Observed | Desired Lead Time |
|---|---|---|
| Surface Wind (including gusts) ≥ 26 kts < 35 kts | Forecast | 30 minutes |
| Surface Wind (including gusts) ≥ 18 kts | Observed | N/A |
| Crosswind ≥ 20 kts | Observed | N/A |
| Fighter Thermal Stress CAUTION | Observed | N/A |
| Fighter Thermal Stress DANGER | Observed | N/A |
| Ice Foreign Object Debris is occurring within 10NM When any one or more of the following criteria exist: | Observed | N/A |
| (1) Ambient temperatures between 20ºF (-7ºC) and 45ºF (7ºC) with precipitation.<br>(2) Dewpoint within 9ºF (5ºC) of ambient temperatures between 25ºF (-4ºC) and 45ºF (7ºC).<br>(3) Ambient temperature below 45ºF (7ºC) with standing water or a mixture of water with ice or snow within the immediate proximity of the engine inlet. | Observed | N/A |
| Sea Condition RED (W470 & W151) is occurring when sustained surface winds are ≥ 25 kts and/or wave heights are ≥ 10 feet | Observed | N/A |
| Sea Condition YELLOW (W470 & W151) is occurring when sustained surface winds are ≥ 20 kts < 25kts and/or wave heights ≥ 8 feet < 10 feet | Observed | N/A |

## 5.5. Watches/Warnings/Advisories Format.

5.5.1. Advisories, watches, and warnings will be numbered consecutively by identifying the type of weather message (watch, warning, or advisory) followed by a five-digit number. The first two numbers indicate the current month while the second three numbers indicate the sequence number. For example, the message "Weather Warning 02-005" means the month is February (02), and this is the fifth (005) warning issued in the month. The message "Weather Advisory 12-013" means the month is December (12), and this is the thirteenth (013) advisory issued in the month.

**Note:** The DAF 3806, *Weather Watch Advisory Log*, and/or DAF 3807, *Watch/Warning Notification and Verification,* will be filled out by the forecaster if JET is inoperable.

**Note:** Lightning watches and observed lightning warnings are separate entities and do not supersede previously issued watches/warnings for other criteria.

**5.6. Upgrades/Downgrades.**

5.6.1. Advisories and warnings will be upgraded (i.e. winds increase from 35 KTS to 50 KTS) or downgraded as required. Upgrades should meet the desired lead-times specified above. If a warning is issued for one criterion, and it becomes necessary to warn for other criteria, a new warning, with a new number, will be issued to include all criteria expected. A separate valid time may be specified for each criterion, if necessary.

**5.7. Amendments.**

5.7.1. Amendments to weather warnings and watches will only be issued to change the valid time and will be issued before the original watch or warning expires. New warnings and watches will be issued for any change in weather criteria.

**5.8. Cancellation.**

5.8.1. Warnings and watches may be cancelled when the weather phenomena are no longer occurring or expected to occur. However, if not cancelled, they will expire at the end of the valid period.

**5.9. Severe Weather Action Procedures (SWAP).**

5.9.1. The WF will have procedures in place to initiate, perform, conclude, and complete post event review IAW DAFMAN 15-129.

5.9.2. SWAP Exercises. Per DAFMAN 15-129 4.12.1 the WF will conduct an annual severe weather response exercise. Real world severe weather events serve the purpose of an exercise.

**Note:** WF operational support may be limited to all base units/customers during severe weather events due to limited manning, availability, and duty priorities.

**5.10. Severe Weather Damage Reporting/Operational Reports (OPREPS).**

5.10.1. IAW DAFMAN 15-129, and AFMAN 10-206, the WF will query base agencies local spotters, and online damage assessment data sources to collect the area impacts of severe weather events as time permits after a severe weather occurrence. If there is damage related to severe weather, the WF will coordinate with the 325 FW/CP for weather-related Operational Report 3 (OPREP-3) and provide 325 FW/CP any pertinent base information. The 325 FW/CP, in turn, will provide the WF with a copy of any weather-related OPREPs. The WF will provide damage reports and OPREP-3s to the OWS and higher headquarters as soon as possible.

5.10.2. The 325 FW/CC is the releasing authority for OPREP-3s. Information required to be provided by the WF to the 325 FW/CP includes:

5.10.2.1. Actual severe weather experienced

5.10.2.2. Forecast valid at the time of occurrence

5.10.2.3. Actual severe weather experienced

5.10.2.4. Any watches or warnings issued (include actual and desired lead times)

5.10.2.5. Operational status of meteorological equipment (radar, FMQ-19, etc.)

5.10.2.6. Any other pertinent information (damage reports, visual/radar tornadic confirmations, etc.)

# Chapter 6

# DISASTER SUPPORT

## 6.1. General.

6.1.1. The WF will provide specialized weather support for major accidents, natural disasters and wartime contingencies. The 26 OWS may also supplement the WF with weather data-only support during closed operational hours or outages if required (reference 26 OWS-Tyndall AFB Installation Data Page).

## 6.2. Chemical Downwind Messages (CDM).

6.2.1. The 325 CES Readiness Flight & Emergency Management Flight is the primary user of CDMs provided via the 26 OWS website or the WF. If the CDMs are not available via the internet, the WF will compute the CDM manually. CDM bulletins are available on the OWS website **https://26ows.us.af.mil/product/cdm**.

## 6.3. Effective Downwind Messages.

6.3.1. The 325 CES Readiness & Emergency Management Flight is the primary user of effective downwind messages. These can be created via AFW-WEBS or on the 26 OWS website **https://26ows.us.af.mil/product/edm/**.

## Chapter 7

## WEATHER INFORMATION DISSEMINATION

### 7.1. General.

7.1.1. The WF has multiple methods of disseminating weather information such as TAFs, observations, and WWAs. Primary and secondary notification methods are described in this chapter.

### 7.2. Dissemination and Back-up Systems.

7.2.1. Joint Environmental Toolkit (JET). The WF uses JET as its primary method of disseminating observations, forecasts, warnings, watches, and advisories. If a unit feels they require JET access, contact WF leadership. Agencies without JET access will receive critical weather information through their unit's own dissemination system. The WF will follow documented back-up dissemination procedures in the event that JET is inoperative.

7.2.2. The Air Force Intranet (AFNET). The AFNET is the primary method used for disseminating web based MWPs. Additionally, it is used as the primary system for disseminating (long- line only) and receiving all weather information via JET.

7.2.3. Hotlines. Hotlines have been installed between the WF and the ATC Tower for the sole purpose of rapidly exchanging weather data. Normal phone lines will be used in the event of a hotline outage.

7.2.4. TAFs. TAFs are disseminated via JET. If JET is out of service, then the WF will disseminate the information locally to the Tower/SOF, and Radar Approach and Control (RAPCON).

7.2.5. Observations Procedure. Observations are disseminated via JET. When JET is out of service, observations will be disseminated long-line or via other applicable back-up procedures. Locally, observations will be relayed by the WF to the following organizations (noted on a local dissemination log), in order of priority when JET is out of service prior to long-line transmission:

7.2.5.1. Tower

7.2.5.2. RAPCON

7.2.5.3. SOF

7.2.5.4. Command Post (as requested)

7.2.5.5. Airfield Management (verbally when the airfield is open, as requested)

7.2.5.6. 26 OWS

**Note:** Other base organizations can view observations through the JET Guest page.

7.2.6. FITS. The WF will disseminate FITS via JET as an observed advisory. Additionally, the WF will give a courtesy call to 325 FW/CP. Direct phone calls will serve as the primary means of dissemination during JET outages.

7.2.7. Weather Watches, Warnings, and Advisories will be disseminated via JET. Further dissemination by Airfield Management and the 325 FW/CP is as follows:

7.2.7.1. Airfield Management. Airfield Management relays all weather warnings via the secondary crash net.

7.2.7.2. Command Post. The 325 FW/CP disseminates all weather watches, warnings, and advisories via AtHoc.

7.2.8. Lightning Warnings. The 325 FW/CP retains responsibility for activating the Giant Voice system upon issuance of lightning warnings.

7.2.9. Severe Weather. When any type of severe weather watch or warning is issued, the 325 FW/CP notifies the base leadership via the Commander's Net.

7.2.10. Tornado Warnings. The 325 FW/CP has the primary responsibility for sounding the base siren in the event of a tornado warning issuance. This will be followed up with activation of the Commander's Net, AtHoc, and the Emergency Cable Override system to provide additional warning to the base populace.

# Chapter 8

# WEATHER EQUIPMENT

## 8.1. General.

8.1.1. The WF uses several different systems to observe and disseminate weather information. Primary and backup systems are described in this chapter.

## 8.2. Meteorological Equipment.

8.2.1. FMQ-19. FMQ-19 is an automated observation sensor system responsible for taking and disseminating all METAR and SPECI observations for the primary airfield complex. The FMQ-19 observes wind speed and direction, altimeter, relative humidity (dew point), air temperature, visibility, cloud height and coverage, precipitation type, precipitation amount, lightning, and Runway Visual Range (RVR). The FMQ-19 is the primary lightning sensor for Tyndall.

8.2.2. FMQ-22. FMQ-22 is an automated observation sensor system responsible for taking and disseminating all METAR and SPECI observations for the Tyndall drone runway. The FMQ-22 system consists of one sensor and can be utilized as a backup lightning resource due to its proximity to the primary FMQ-19 system.

8.2.3. TMQ-53. TMQ-53 is a deployable automated observation sensor system. The TMQ- 53 is capable of recording METAR and SPECI observations similar to the FMQ-19/22. The TMQ-53 is used in deployed/contingency environments and can serve as a backup should the FMQ-19/22 becomes unserviceable.

8.2.4. M625 Micro Weather Sensor is a deployable automated observation sensor system. It is capable of recording METAR and SPECI observations similar to the FMQ-19/22.

8.2.5. Kestrel. Kestrels are handheld weather sensors utilized for manual weather observations. Kestrels can be used for wind speed and direction, altimeter, relative humidity, and air temperature.

## 8.3. Communications Equipment.

8.3.1. JET. This is the primary system for disseminating forecasts, observations, PIREPS, DD175-1 briefings, warnings, watches, and advisories. JET disseminates all information via the AFNET.

8.3.2. Phones/Hotlines. Phones and hotlines primarily serve as back-up system and are also used to pass along critical, time-sensitive information rapidly.

8.3.3. AFNET. The WF relies heavily on the Air Force Network to improve the timeliness and accuracy of weather intelligence to our supported units.

## 8.4. Maintenance.

8.4.1. The following organizations provide preventive maintenance and repair weather and communications equipment:

**Table 8.1. Weather Equipment Maintenance.**

| Organization | Equipment |
|---|---|
| 325 OSS/OSM (RAWS) | FMQ-19 (Primary Airfield Complex) |
| 325 OSS/OSM | FMQ-22 (Droneway) |
| 325 OSS/OSM | TMQ-53 |
| 557 Weather Wing (WW) | GIBSON RIDGE Analyst |
| Base Network Control Center/JET Help Desk | JET |
| 325 CS | Phones/Hotlines |

# Chapter 9

# RECIPROCAL SUPPORT

## 9.1. General.

9.1.1. This chapter describes support required by the WF in order to accomplish its daily mission. Per DAFMAN 15-129, support to the WF by other local agencies that are mandated by USAF or other local directives are not included in this chapter.

## 9.2. 325 CS will:

9.2.1. Coordinate with the 325 OSS if on-site support is required for physical access to the server.

9.2.2. Be notified of any significant changes from the 325 OSS and JET helpdesk from updates on all applicable technical orders.

## 9.3. 325 OSS/OSM will:

9.3.1. Provide, coordinate, or arrange for the installation, maintenance, and repair of all weather communication and meteorological sensing equipment, except for the contract-maintained JET hardware.

9.3.2. Maintain and update all technical orders and will advise operators of any significant changes, as received.

9.3.3. As the single point of contact, RAWS will:

9.3.3.1. Issue job control numbers for all outages.

9.3.3.2. Notify the responsible service agents for outages.

9.3.3.3. Coordinate with off-base agencies to repair off-base lines.

9.3.3.4. Perform necessary follow-up actions as required until full service is restored.

9.3.3.5. Ensure that established maintenance response times are met.

9.3.3.6. Ensure a 24-hour point of contact for reporting outages and assigning Job Control numbers is available.

9.3.3.7. Coordinate with Tyndall WF prior to taking any equipment down for scheduled maintenance.

## 9.4. 325 OSS/OSAA will:

9.4.1. Disseminate weather watches and warnings over the ramp net and weather warnings over the secondary crash net.

9.4.2. Notify WF personnel of any aircraft mishaps or in-flight emergencies.

9.4.3. Notify WF personnel of all impending changes to any flight information publications (FLIP) affecting Tyndall weather support requirements.

9.4.4. Forward WF initiated changes of flight information publications and chart orders to the appropriate agencies for action.

9.4.5. Advise WF leadership if the airfield will be open or closed for any period of time outside of the published airfield operating hours.

**9.5. 325 OSS/OSAR will:**

9.5.1. Relay all PIREPS to the WF not later than 5 minutes after receipt.

**9.6. 325 OSS/OSAT will:**

9.6.1. Provide tower orientation to new WF personnel.

9.6.2. Participate in the cooperative weather watch program as outlined in section 2.4.

9.6.3. Notify the WF of all active runway changes when complete.

9.6.4. Receive local weather phenomena training and prevailing visibility certification.

9.6.5. Relay all PIREPS to the WF no later than 5 minutes after receipt.

9.6.6. Notify the WF of any Airfield Automation System outages.

**9.7. 325 OSS/OSO will:**

9.7.1. Ensure current flying schedule is available to the WF.

**9.8. 325 FW/CP will:**

9.8.1. Disseminate weather warnings, watches, and advisories according to local procedures.

9.8.2. Notify senior wing leadership when any weather warning or severe weather watch (e.g., tornado, winds $\geq$ 50 KTS, or $\geq$ ¾-inch hail) is issued.

9.8.3. Notify the WF when any other agency reports a funnel cloud, waterspout, tornado, or any other significant weather event.

9.8.4. Notify the WF of any significant weather related event (material damage, injuries, etc.) as time permits and conditions dictate.

**9.9. 325 SFS will:**

9.9.1. Report the sighting of any waterspout, tornado, funnel cloud, hail, or freezing precipitation to the WF directly or via the 325 FW/CP.

**9.10. 325 TRSS, 2 FTS, 43 FS, 95 FS will:**

9.10.1. Provide PIREPS, either directly to the WF, Tower, SOF, or RAPCON.

9.10.2. When possible, provide feedback to the WF for accuracy and timeliness.

9.10.3. Provide two weeks notice if requesting a dedicated forecaster for exercise support (e.g. SENTRY SAVANNAH).

9.10.4. Provide 48 hours notice for any in-person weather briefings (e.g. Instrument Refresher Course, semi-annual weather briefing).

9.10.5. Provide statistics on missions cancelled by weather at KPAM, and/or missions that were forecasted to have mission-limiting/cancelling weather but ended up being flown due to favorable weather.

**9.11. The Supervisor of Flying (SOF) will:**

9.11.1. Notify the weather station if alternate weather is required.

9.11.2. Notify the weather station when an alternate is no longer needed.

9.11.3. Relay all PIREPS no later than 5 minutes after receipt to the WF forecaster.

**9.12. 337 ACS will:**

9.12.1. Provide guidance (at least 2 weeks in advance) to the WF regarding any weather training/educational requirements (or changes in requirements), if applicable.

**9.13. 53 WEG will:**

9.13.1. Provide PIREPS, either directly to the WF, Tower, SOF, or RAPCON.

9.13.2. When possible, provide feedback to the WF for accuracy and timeliness.

9.13.3. Ensure current flying schedule is available to the WF.

9.13.4. Notify WF leadership when any WSEP units will be bringing weather forecasters for exercise support.

9.13.4.1. If home station weather flights are unable to support their aligned units, TAFB weather flight will support given two weeks notice. Subject to staffing availability.

9.13.5. Provide statistics on missions cancelled by weather at KPAM, and/or missions that were forecasted to have mission-limiting/cancelling weather, but ended up being flown due to favorable weather.

**9.14. 325 OMRS/Bioenvironmental will:**

9.14.1. Be the installation point of contact for collecting and disseminating all Wet Bulb Globe Temperature readings and flag conditions (i.e. Green Flag, Black Flag).

**9.15. All Weather Support Recipients will:**

9.15.1. Notify WF of problems with JET.

9.15.2. Notify WF through proper chain of command when new weather support requirements are identified.

9.15.3. Coordinate changes/additions to weather support requirements as soon as they are foreseen.

9.15.4. Provide a minimum of two weeks notice for known weather support requests requiring out-of-station support.

9.15.5. When possible, provide feedback to the WF for accuracy and timeliness.

# Chapter 10

# TROPICAL SUPPORT

## 10.1. General.

10.1.1. Atlantic tropical season runs from 1 June through 30 November, but tropical systems can threaten Tyndall any time of the year. The National Hurricane Center (NHC) advisories, official forecast information, are issued every 6 hours and accessible via the NHC website. The WF will not deviate from the broad overarching official forecast but have leeway to tailor Tyndall forecast products with special consideration given to local terrain effects (over- land wind fetch, bay effects, etc.).

## 10.2. Tropical Bulletin.

10.2.1. The WF will tailor the 26 OWS Tropical Cyclone Threat Assessment Product (TC-TAP) and NHC advisories to issue Tyndall specific tropical bulletins to alert base leadership, personnel, and agencies of the potential impacts at Tyndall or if a tropical system is within 400 NM of Tyndall.

10.2.2. Tropical Bulletin Dissemination. The bulletin will be emailed to the 325 FW and associate unit commanders, and to other base leadership for coordination. To be added to the WF Tropical/Severe Weather distribution list, please contact WF leadership.

**Note:** The 26 OWS will issue tropical effects watches and warnings as described in section 5.5.

## 10.3. Hurricane Condition (HURCON).

10.3.1. HURCON definitions for TAFB are established in TAFB Installation Emergency Management Plan (IEMP) 10-2.

## 10.4. HURCON Crisis Action Team /Emergency Operations Center Meetings.

10.4.1. When the 325 FW commander calls a HURCON meeting, the WF will brief the current and forecast track/intensity of the storm, wind/storm surge impacts on Tyndall, and a planning timeline based on the latest NHC forecast. Emergency Operations Center briefs occur as requested. Subsequent Crisis Action Team or Emergency Operations Center meetings are scheduled as necessary.

10.4.2. Tropical impacts expected at Eglin AFB will be obtained from the 96 WS.

**10.5. Hurricane Base Recovery Element (BRE) and Ride Out Element (ROE).**

10.5.1. BRE and ROE members from the WF will be dictated by the current TAFB IEMP 10-2.

10.5.2. In the event the 325 FW/CC orders a base evacuation, WF members not on the BRE and ROE will evacuate. In order to ensure that there are no lapses in personnel available to brief tropical updates, the WF will, with prior coordination with 325 OSS & 325 OG leadership, preempt the evacuation order and release one forecaster early so that WF operations can continue while the remaining personnel evacuate.

CHRISTIAN M. BERGTHOLDT, Col, USAF
Commander

## Attachment 1

## GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

***References***

ACCMAN 11-2E-9V3, *E-9--Operations Procedures*, 24 July 2022

ACCMAN 11-2QF-16V3, *QF-16--Operations Procedures*, 26 March 2024

AFI 11-418, *Operations Supervision*, 21 December 2021

AFI 33-322, *Records Management and Information Governance Program,* 27 July 2021

AFMAN 10-206, *Operational Reporting*, 31 August 2020, ACC Supp, 5 October 2022

AFMAN 11-202V3, *Flight Operations*, 09 January 2022, ACC Supp, 7 November 2022

AFMAN 11-210, *Instrument Refresher Program (IRP),* 21 December 2021

AFMAN 15-111, *Surface Weather Observations*, 11 March 2019

AFMAN 15-124*, Meteorological Codes*, 15 January 2019

AFPD 15-1*, Weather Operations*, 13 November 2019

DAFI 10-208, *Continuity of Operations (COOP) Program,* 5 February 2024, ACC Supp, 1 November 2023

DAFI 10-2501, *Emergency Management Program*, 15 October 2023

DAFI 48-151, *Thermal Stress Program,* 1 May 2022

DAFMAN 15-129, *Air and Space Weather Operations,* 06 September 2023

TAFB Installation Emergency Management Plan (IEMP) 10-2, 1 April 2022

TYNDALLAFBI 11-250, *Air Operations Procedures and Programs*, 13 April 2022

***Prescribed Forms***

None

***Adopted Forms***

DAF 3806, *Weather Watch* Advisory Log

DAF 3807, *Watch/Warning Notification and Verification*

DAF Form 847, *Recommendation for Change of Publication*

DD Form 175-1, *Flight Weather Briefing*

***Abbreviations and Acronyms***

**26 OWS**—26th Operational Weather Squadron

**ACC**—Air Combat Command

**AF**—Air Force

**AFB**—Air Force Base

**AFI**—Air Force Instruction

**AFMAN**—Air Force Manual

**AFNET**—Air Force Network

**ATC**—Air Traffic Control

**CDM**—Chemical Downwind Message

**CS**—Communications Squadron

**FBWOS**—Fixed Base Weather Observing System

**FITS**—Fighter Index of Thermal Stress

**FLIP**—Flight Information Publication

**FMQ**—Fixed Meteorological Equipment

**FW**—Fighter Wing

**HURCON**—Hurricane Condition

**IAW**—In Accordance With

**IEMP**—Installation Emergency Management Plan

**JET**—Joint Environmental Toolkit

**KTS**—Knots

**LAN**—Local Area Network

**METAR**—Routine Meteorological Observation Report

**METWATCH**—Meteorological Watch

**MISSIONWATCH**—Mission-Scale Meteorological Watch

**MOA**—Military Operating Area

**MWP**—Mission Weather Product

**NHC**—National Hurricane Center

**OG**—Operations Group

**OPR**—Office of Primary Responsibility

**OPREP**—Operational Report

**OSS**—Operations Support Squadron

**OWS**—Operational Weather Squadron

**PIREP**—Pilot Report

**PMSV**—Pilot-to-Metro Service

**RAPCON**—Radar Approach and Control

**RAWS**—Radar Airfield & Weather Systems

**RVR**—Runway Visual Range

**RVRNO**—Runway Visual Range Information Not Available

**SOF**—Supervisor of Flying

**SPECI**—Special

**SWAP**—Severe Weather Action Procedures

**TAF**—Terminal Aerodrome Forecast

**TAFB**—Tyndall Air Force Base

**TC-TAP**—Tropical Cyclone Threat Assessment Product

**USAF**—United States Air Force

**VIS**—Visibility

**WEG**—Weapons Evaluation Group

**WF**—Weather Flight

**WW**—Weather Wing

**WWA**—Watch, Warning and Advisory

## Attachment 2

## CUSTOMER WEATHER LIMITATIONS AND ACTIONS

**Table A2.1. Customer Weather Limitations and Actions.**

| **F-35 Flying Operations** | | |
|---|---|---|
| **WEATHER CONDITION** | **CUSTOMER ACTION** | **IMPACT** |
| Wet runway Crosswinds >20 KTS | Weather hold/divert/reschedule | Delay/Cancel sorties; Divert aircraft airborne |
| Dry runway Crosswinds (no limitations) | None | None |
| Winds > 35 KTS steady state | Weather hold/divert/reschedule | Delay/Cancel sorties; Divert aircraft airborne |
| Seas > 10 ft or >25 KTS surface over water | OG waiver for over water flights | Impedes SAR |
| Sea Temp <= 60F | Anti Exposure Suits required | None |
| Ceiling/vis < 500/2 | OG Waiver required for recovery of wingman qualified pilots | Divert if waiver not authorized |
| Icing observed | Alternate MOA/Range | Limits area for flights depending on severity and location |
| Lightning within 35 miles of airfield | Alternate MOA/Range | Limits area for flights |
| Lightning in or within 25 miles of airspace | Alternate MOA/Range or adjust profile to remain >25NM from known lightning | Limits area for flights depending on location |
| Lightning within 25 miles of airfield | Weather hold/divert/delay launch | Delay/Cancel sorties; Divert aircraft airborne |
| Observed within 10 miles of airfield | Weather hold/divert/delay launch | Delay/Cancel sorties; Divert aircraft airborne |
| Lightning within 5 miles of airfield | Weather hold/divert/delay launch | Delay/Cancel sorties; Divert aircraft airborne |
| **Flightline Refueling Operations** | | |
| **Weather Condition** | **Customer Action** | **Impact** |
| Lightning within 5 miles of airfield | Cease aircraft fuel servicing | Delays or cancels sorties |

## Attachment 3

## 325 OSS/OSW DUTY PRIORITIES

**Table A3.1. 325 OSS/OSW Duty Priorities.**

| | |
|---|---|
| 1 | Perform Emergency War Order Tasks (e.g., Deploy Personnel) |
| 2 | Execute Evacuation/Continuity of Operations (COOP) Procedures |
| 3 | Issue/Disseminate Imminent Hazardous Weather Warnings / SWAP |
| 4 | Respond to Aircraft/Ground Emergencies |
| 5 | Issue/Disseminate Imminent Weather Advisories |
| 6 | Respond to PMSV/Phone Patch |
| 7 | Provide Weather Information for Supervisor of Flying (SOF) |
| 8 | Disseminate Terminal Aerodrome Forecasts |
| 9 | Disseminate Urgent (UUA) PIREPS |
| 10 | Provide Flight WX Briefings |
| 11 | Augment FBWOS (FMQ-19) for Mandatory Elements |
| 12 | Collaborate Weather Products with Supported Units |
| 13 | METWATCH/Amend WX Characterization Products |
| 14 | Perform MISSIONWATCH |
| 15 | Respond to Support Assistance Request (SAR) or Request for Information |
| 16 | Provide Staff Briefings / Non-Standard Weather Products |
| 17 | Accomplish Weather Functional Training |
| 18 | Accomplish Administrative Tasks |

## Attachment 4

## SPECIAL OBSERVATION CRITERIA

**A4.1. Criteria.** When the phenomenon tracked in Tables A4.1, A4.2, and A4.3 decreases to or less than, or if below, increases to equal or exceed any of the reportable values. When prevailing visibility is 1 mile or less and the active runway RVR equipment is not in service, or not available, "RVRNO" will be reported. All other observation elements will utilize the SPECI guidance established in AFMAN 15-111.

**A4.2. Single-Element SPECI.** A single-element SPECI is taken only when a delay in reporting all elements would cause an immediate threat to life or property. At Tyndall, a single-element SPECI is authorized for Tornadic Activity and Volcanic Ash.

**Table A4.1. SPECI Ceiling Criteria.**

| Height | Type of Observation | Height | Type of Observation |
|---|---|---|---|
| 5000 feet | Special (53 WEG) | 1000 feet | Special (AFMAN) |
| 3100 feet | Special (325 FW) | 800 feet | Special (AFMAN) |
| 3000 feet | Special (AFMAN) | 700 feet | Special (FLIP) |
| 2600 feet | Special (325 FW) | 600 feet | Special (FLIP) |
| 2500 feet | Special (53 WEG) | 500 feet | Special (FLIP) |
| 2100 feet | Special (325 FW) | 300 feet | Special (325 FW) |
| 2000 feet | Special (FLIP) | 200 feet | Special (FLIP) |
| 1500 feet | Special (AFMAN) | | |

**Table A4.2. SPECI Visibility Criteria.**

| Visibility | Type of Observation | Visibility | Type of Observation |
|---|---|---|---|
| 5 miles | Special (53 WEG) | 1 mile | Special (AFMAN) |
| 3 miles | Special (AFMAN) | ⅞ mile | Special (FLIP) |
| 2 ¼ miles | Special (FLIP) | ¾ mile | Special (FLIP) |
| 2 miles | Special (AFMAN) | ½ mile | Special (FLIP) |
| 1 ½ miles | Special (FLIP) | | |
| 1 ⅜ miles | Special (FLIP) | | |

**Table A4.3. RVR Visibility Criteria.**

| RVR | Type of Observation | RVR | Type of Observation |
|---|---|---|---|
| 6000 feet | Special (AFMAN) | 4000 feet | Special (FLIP) |
| 5000 feet | Special (AFMAN) | 2400 feet | Special (AFMAN) |
| 4500 feet | Special (FLIP) | 2000 feet | Special (AFMAN) |

## Attachment 5

## TAF SPECIFCATION AND AMENDMENT CRITERIA

**A5.1. TAF Specification and Amendment Criteria.** The TAF will be amended at the time of occurrence, the duration, and the intensity (if applicable) of expected amendment weather criteria as listed in **Table A5.1** for ceilings and visibility & DAFMAN 15-129 Table 5.3.

**Table A5.1. Ceiling and Visibility Specification and Amendment Criteria.**

| Ceiling | Visibility | Category | Source |
|---|---|---|---|
| ≥ 2500 FT* | ≥ 5SM | F | AFI 11-202 V3 ACC Sup I |
| < 2500 FT but ≥ 2000 FT | ≥ 3SM but < 5SM | E | DFMAN 15-129 |
| < 2000 FT but ≥ 1000 FT | < 3SM but ≥ 2SM | D | DFMAN 15-129 |
| < 1000 FT but ≥ 700 FT | < 3SM but ≥ 2SM | C | DFMAN 15-129 |
| < 700 FT but ≥ 200 FT | < 2 SM but ≥ 1/2SM | B | AFI 11-202 V3 |
| < 200 FT | < 1/2SM | A | AFI 11-202 V3 |
| 5000 FT | N/A | Specification | AFI 11-202 V3 ACC Sup I |
| 1500 FT | N/A | Specification | AFI 11-202 V3 ACC Sup I |
| 500 FT | 1 1/2SM | Specification | AFI 11-202 V3 ACC Sup I |
| 300 FT | 1 SM | Specification | AFI 11-202 V3 ACC Sup I |

***When observed on either the KTDR or KPAM sensors.**